



**XMBA 200S: Data and Decisions**

Chapter 20: Curved Patterns

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# Course overview

## Part 4: Building regression models

- Block 3 and 4
- You want to build a model to forecast outcomes from data you have
- What's the best **model you can build**?
- Today: constructing and interpreting the OLS regression line
  - Future classes: inference; multivariate models

# Outline

- Modeling and Interpreting Curved Patterns
- Transformations

## Quick Poll

Consider a manager whose salespeople drive a fleet of cars, each driving 25,000 miles per year.

In upgrading the fleet, what has more of an impact on fuel consumption?

- Replacing a 25MPG car with 35MPG car
- Replacing a 12MPG car with 14MPG car

## Quick Poll

The answer is (b). Why? Calculate the amount of gallons used per year for each vehicle.

|          |                            |                         |
|----------|----------------------------|-------------------------|
| 25 mpg → | 1,000 gallons              | (25,000 miles / 25 mpg) |
| 35 mpg → | 714 gallons                | (25,000 miles / 35 mpg) |
|          | -----                      |                         |
|          | 286 gallons saved per year |                         |

|          |                            |                         |
|----------|----------------------------|-------------------------|
| 12 mpg → | 2,083 gallons              | (25,000 miles / 12 mpg) |
| 14 mpg → | 1,786 gallons              | (25,000 miles / 14 mpg) |
|          | -----                      |                         |
|          | 297 gallons saved per year |                         |

# Detecting Nonlinear Patterns

## Vehicle Weight and Fuel Efficiency

What is the relationship?

What is the average fuel efficiency of a vehicle that weighs 3000 pounds?

On average, how much does fuel efficiency decrease for a 200 pound increase in vehicle weight?

Is this relationship the same at different weight levels?

# Detecting Nonlinear Patterns

## Vehicle Weight and Fuel Efficiency

You have data on 303 vehicle models marketed in the United States. For each vehicle you observe the *MPG* and the *Weight*, measured in thousands of pounds.

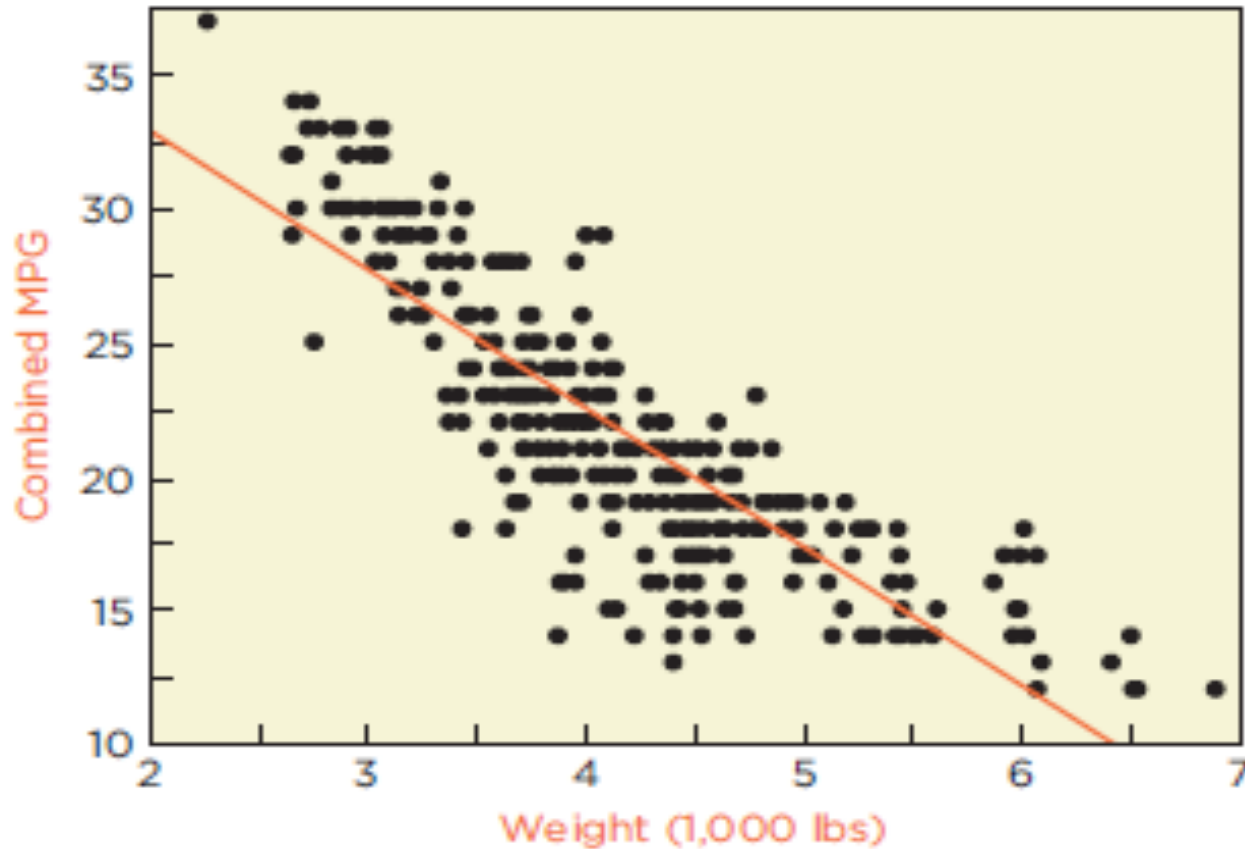
You calculate that the least squares fitted line is,

$$\text{Predicted MPG} = 43.3 - 5.19 \text{ Weight.}$$

The line has an  $r^2 = 0.70$  and  $s_e = 2.9$  MPG.

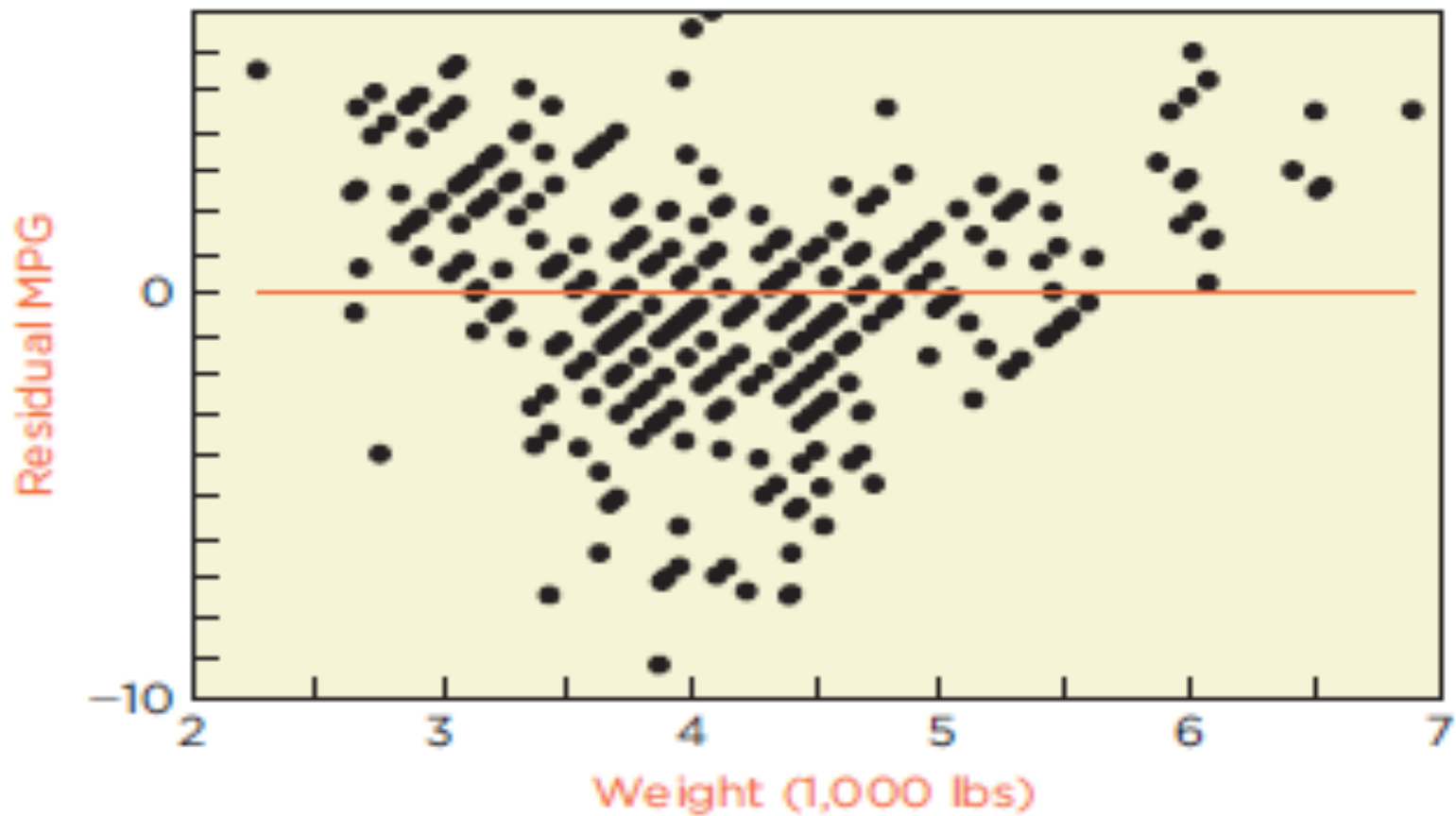
# Detecting Nonlinear Patterns

## Scatterplot with Fitted Line



# Detecting Nonlinear Patterns

## Residual Plot

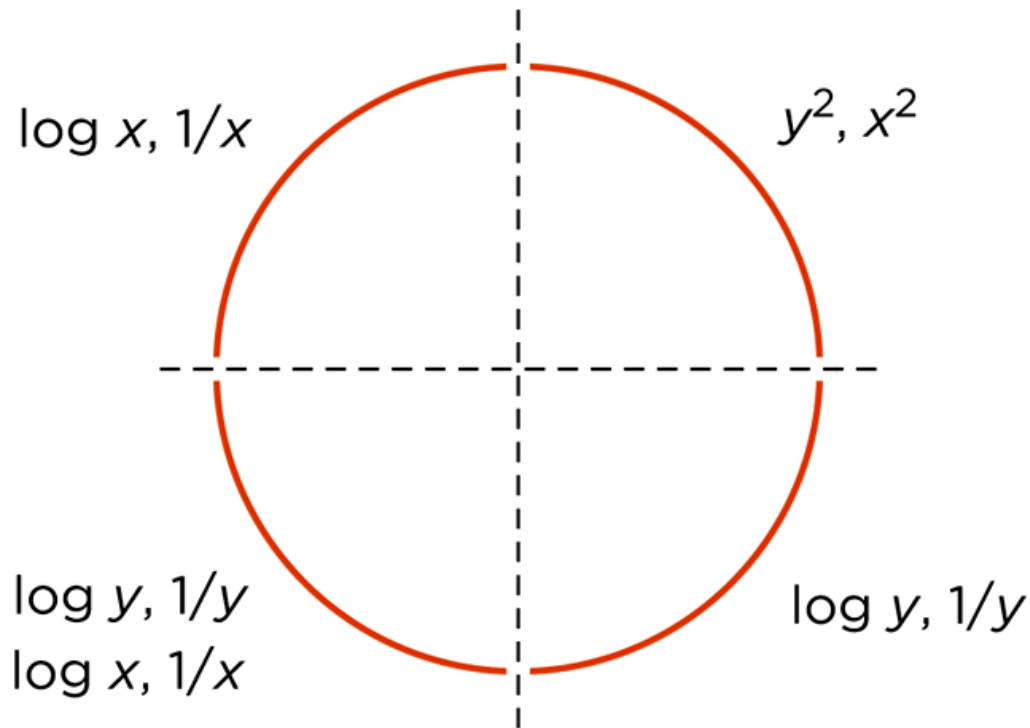


# Detecting Nonlinear Patterns

## Transformations

- A transformation is the re-expression of a variable by applying a function to each observation.
- Transformations allow the use of regression analysis to describe a curved pattern.
- Two nonlinear transformations that are particularly useful in business applications are the reciprocal and logarithms.

# Choosing a Transformation



Match the pattern in a scatterplot of  $y$  on  $x$  to one of the shapes to find an appropriate transformation.

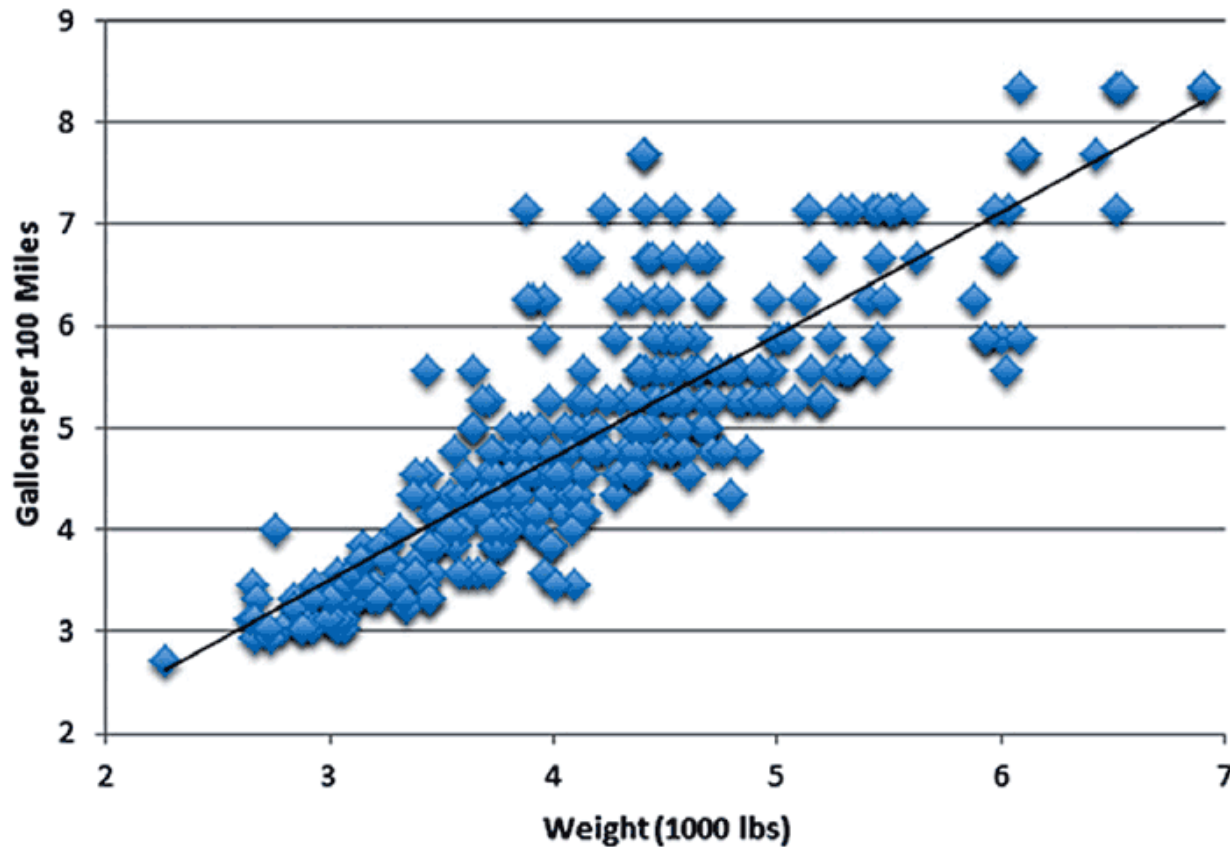
# Reciprocal Transformation

## Ratios

- The reciprocal transformation is useful for variables that are in the form of a ratio, such as miles per gallon.
- Other examples: equity/debt, sales/employee, price/assets.
- In the vehicle weight example, apply this transformation to fuel efficiency (miles per gallon) and multiply by 100.
  - The resulting response is number of gallons it takes to go 100 miles.

# Reciprocal Transformation

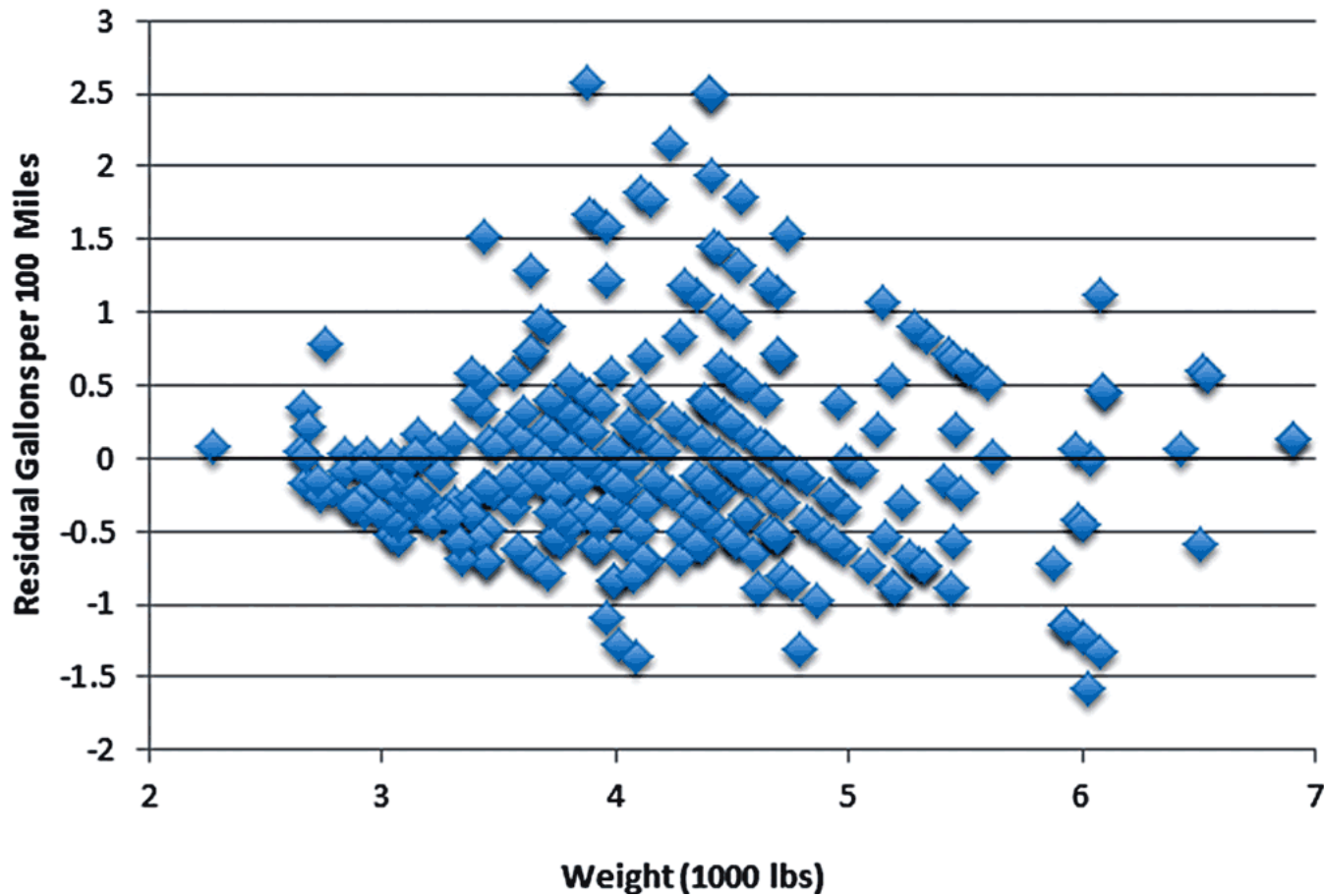
## Scatterplot with Fitted Line



$$\text{Gallons Per 100 Miles} = -0.11 + 1.20 \text{ Weight} \quad r^2 = 0.71 \quad se = 0.667$$

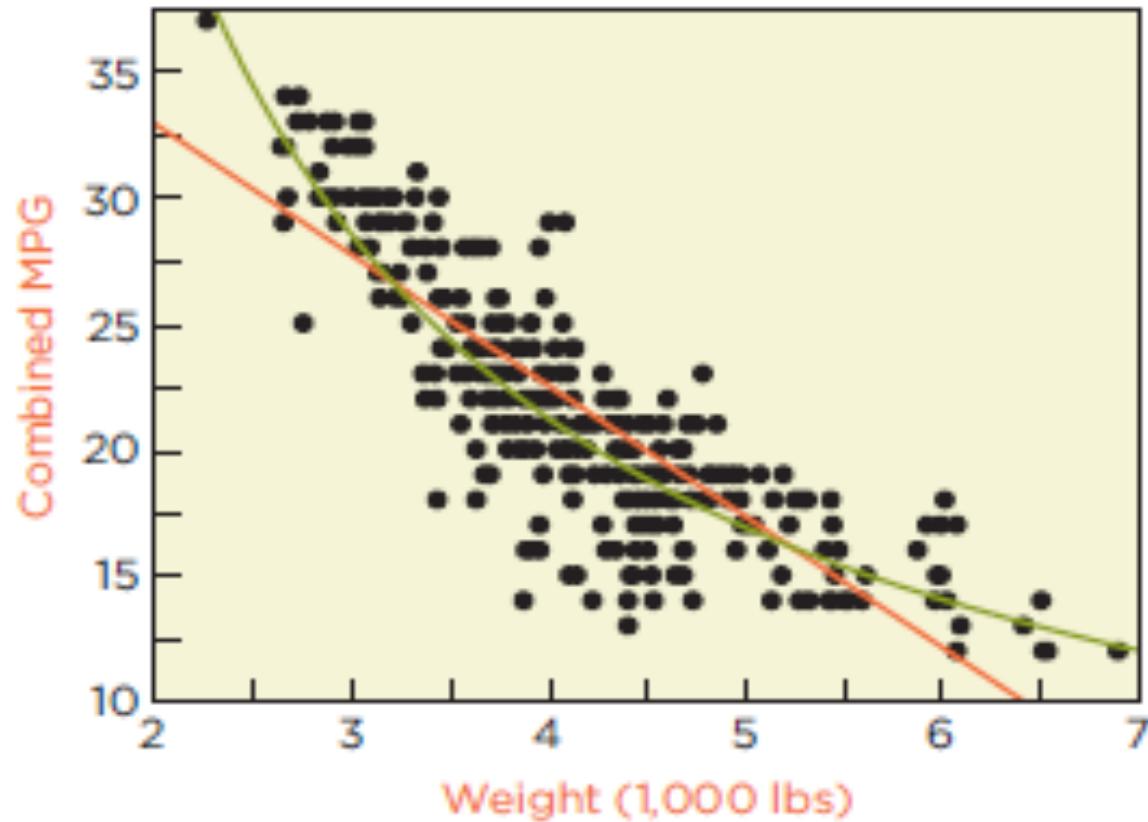
# Reciprocal Transformation

## Residual Plot



# Reciprocal Transformation

## Visual Comparison



# Reciprocal Transformation

## Vehicle Weight and Fuel Consumption

On average, how much does fuel consumption increase for a 200 pound increase in vehicle weight?

$$\text{Predicted Gallons Per 100 Miles} = -0.11 + 1.20 \text{ Weight}$$

Multiplying the slope by 0.2 yields 0.24. Thus a 200 pound increase in vehicle weight increases fuel consumption by 0.24 gallons per 100 miles.

This relationship is the same at all weight levels.

# The Incredible Logarithm

If you have a nonlinear relationship, one of the most useful tools is the logarithm.

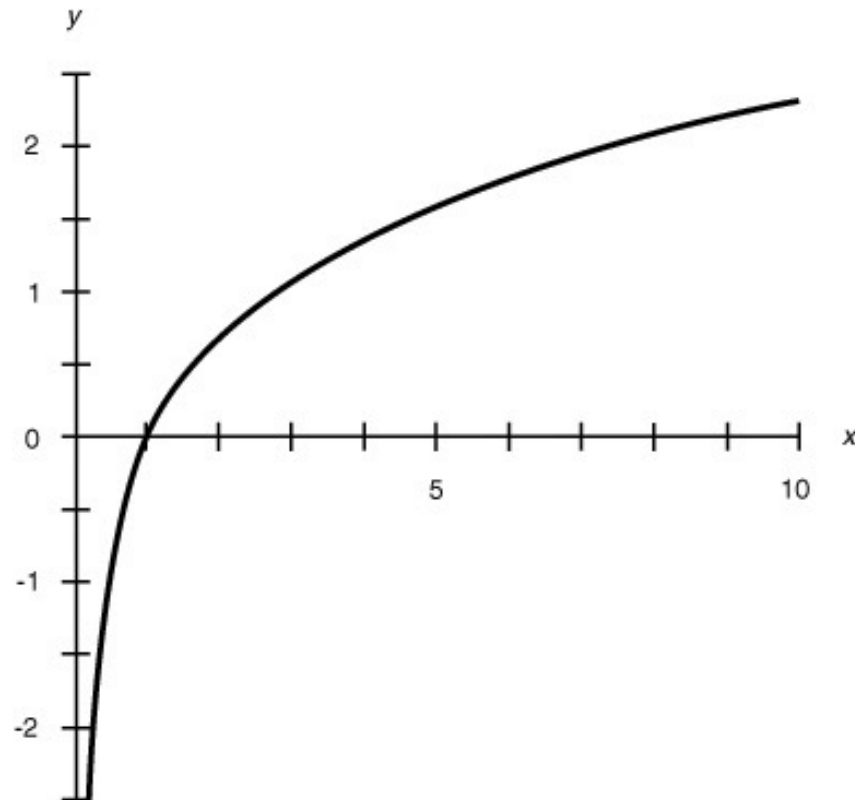
Changes in logs are approximately equal to percentages changes, making logs easy to interpret.

Log models are also much less sensitive to outliers.

We can estimate regression models in which the response variable, explanatory variable, or both, are in logs.

# Log Transformation

## The Log Transformation



# Log Transformation

## Logs and Exponents

Business applications almost always use the “natural” log, which means that the base is euler’s number approximately equal to 2.7.

The natural log of  $x$  is the exponent to which euler’s number must be raised to produce  $x$ .

For example,

$$\begin{aligned}\log(1) &= 0. \text{ Why? Because } 2.7^0 = 1. \\ \log(2.7) &= 1. \text{ Why? Because } 2.7^1 = 2.7. \\ \log(10) &\cong 2.3. \text{ Why? Because } 2.7^{2.3} \cong 10.\end{aligned}$$

# Log Transformation

## Percentage Changes

Changes in natural logs are approximately equal to percentage changes.

For example,

What is the percentage change between 10 and 11? 10%.

What is the change between  $\log(10)$  and  $\log(11)$ ? 0.1.

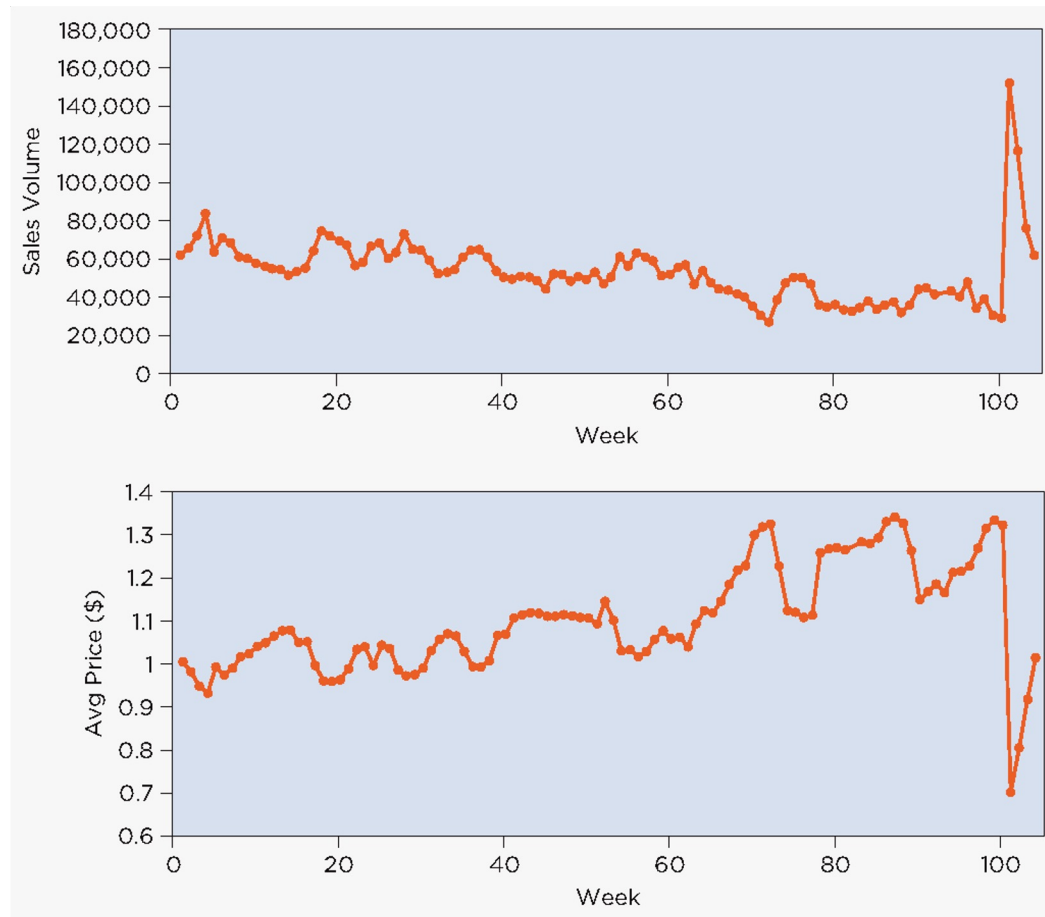
What is the percentage change between 100 and 110? 10%.

What is the change between  $\log(100)$  and  $\log(110)$ ? 0.1.

| x    | log(x) |
|------|--------|
| 1    | 0.0    |
| 10   | 2.3    |
| 11   | 2.4    |
| 100  | 4.6    |
| 110  | 4.7    |
| 1000 | 6.9    |
| 1100 | 7.0    |

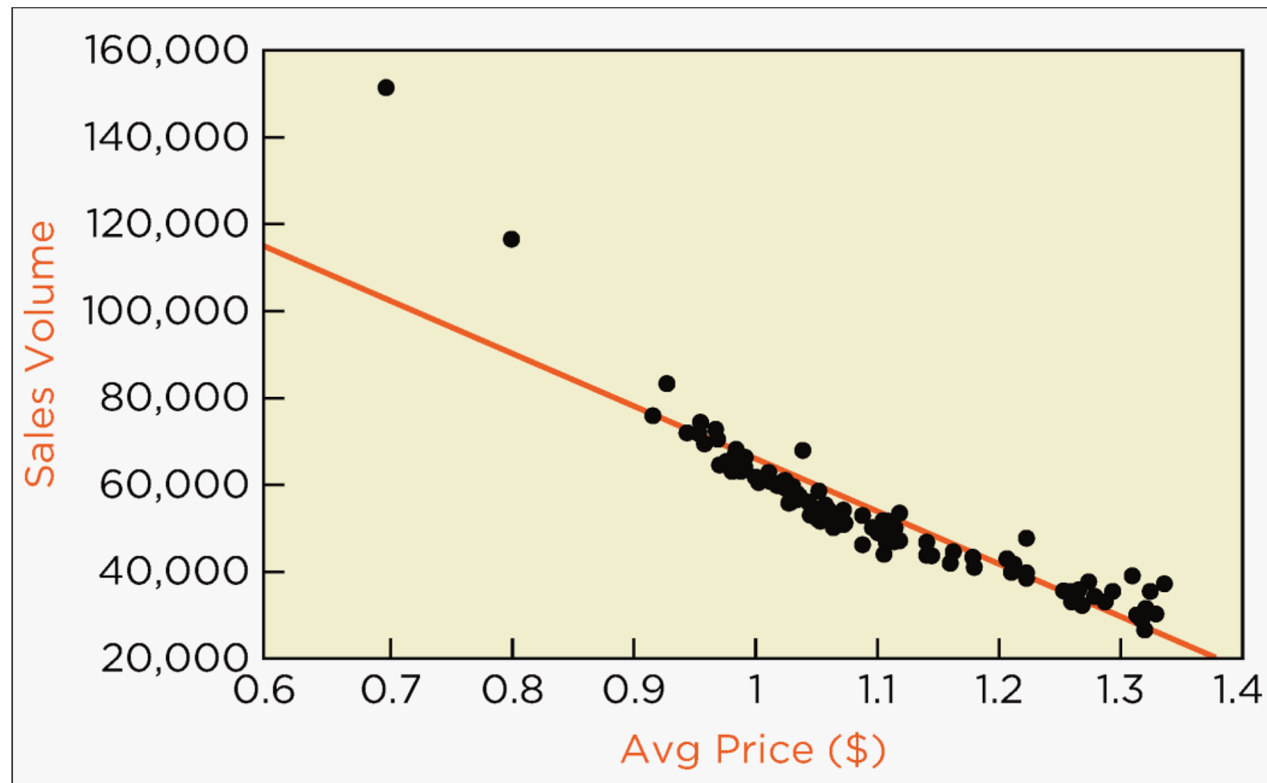
# Log Transformation

## Pet Food Example



# Log Transformation

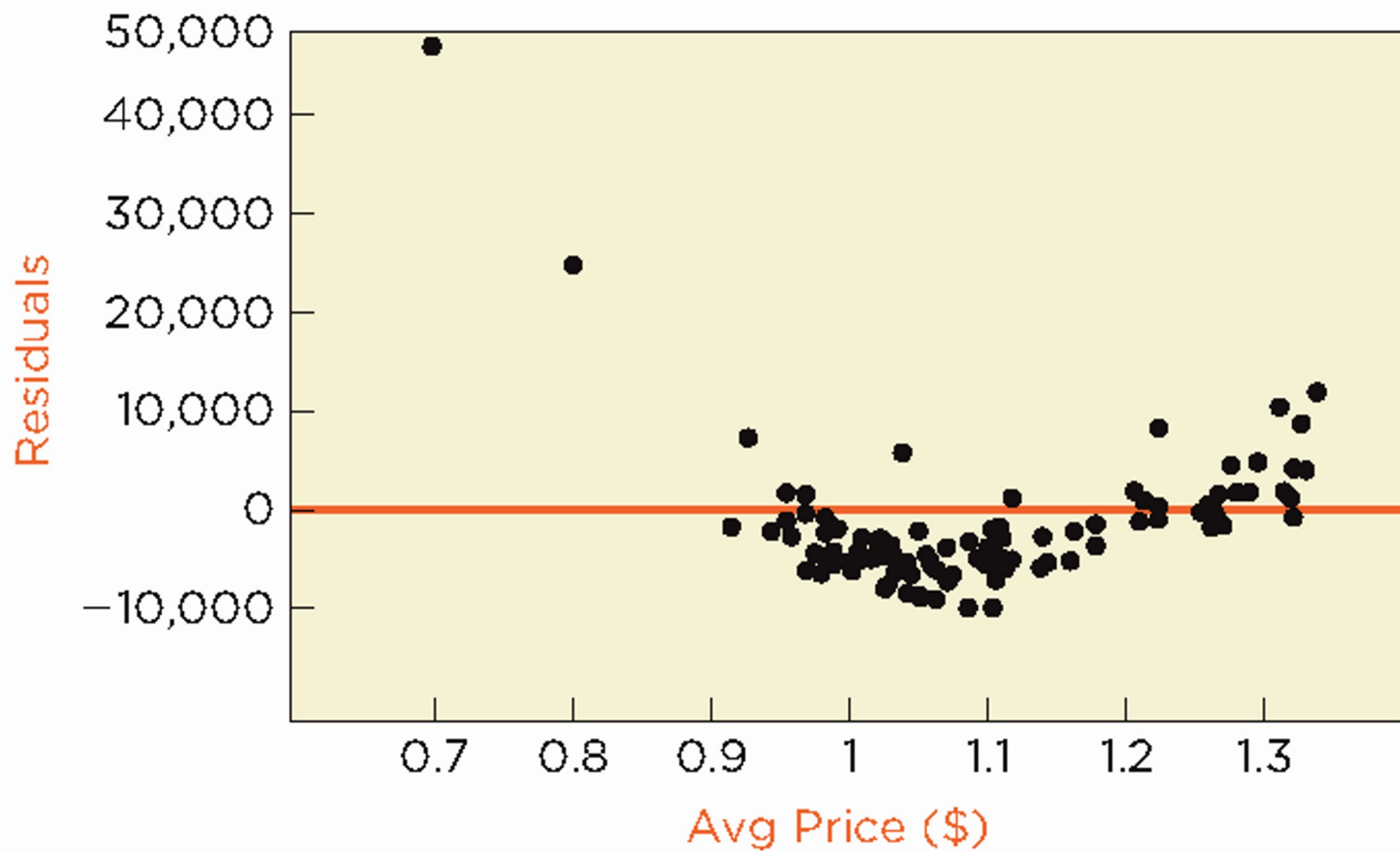
## Scatterplot with Fitted Line



$$\text{Sales Volume} = 190,480 - 125,190 \text{ Price} \quad r^2 = 0.83$$

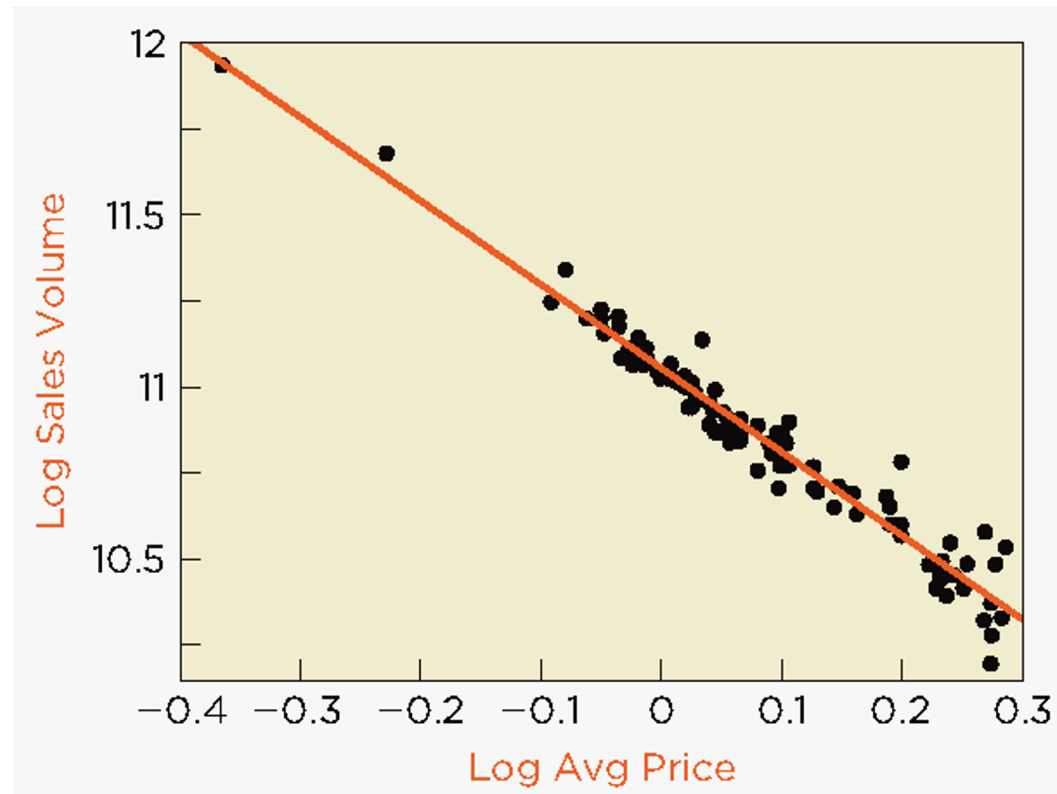
# Log Transformation

## Residual Plot



# Log Transformation

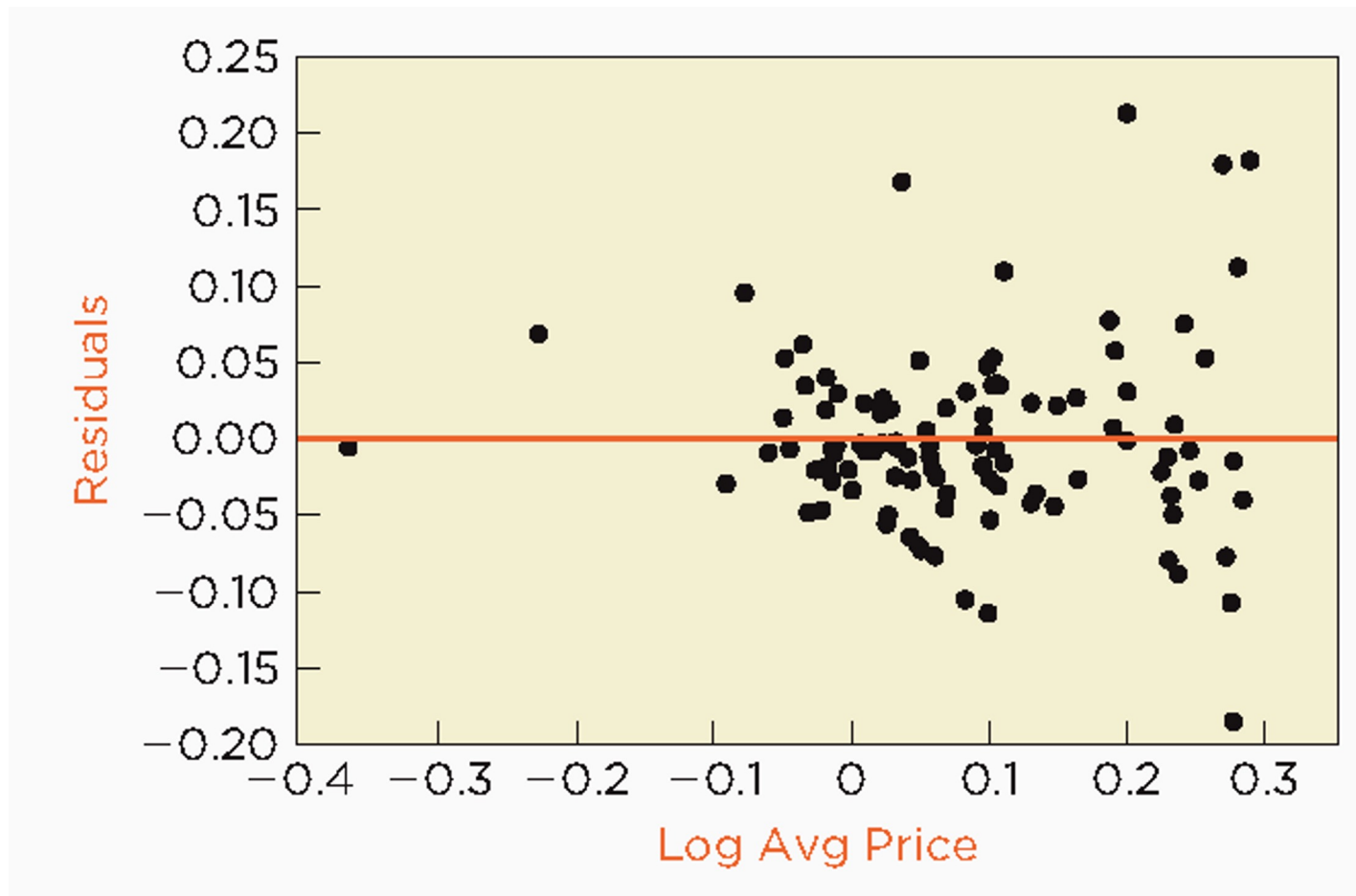
## Scatterplot after Log Transformation



$$\text{Log Sales Volume} = 11.0 - 2.44 \text{ Log Price} \quad r^2 = 0.96$$

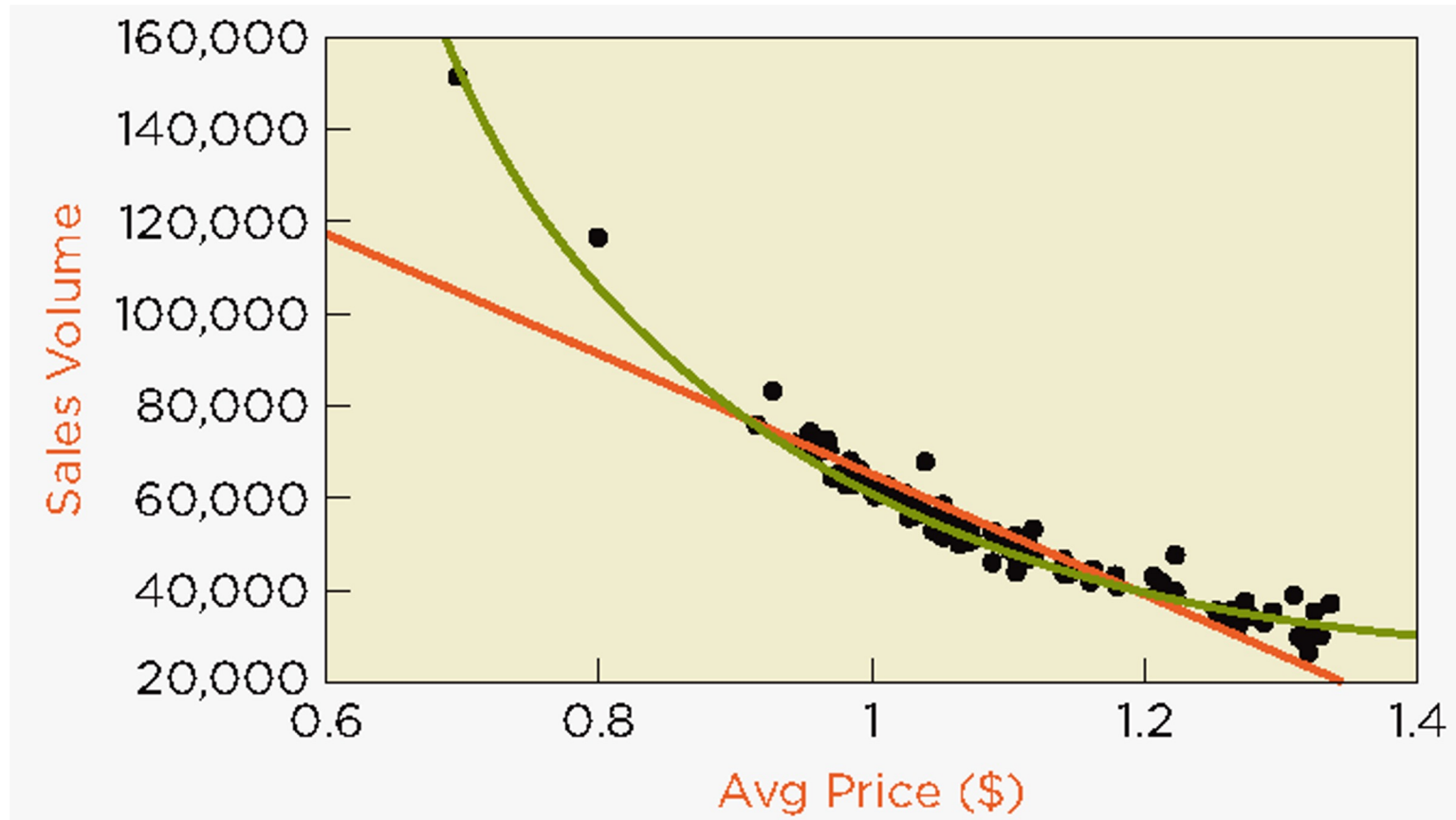
# Log Transformation

## Residual Plot after Log Transformation



# Log Transformation

## Comparing Equations



# Interpreting Logs

## Elasticity

- When both variables are in logs, the estimated slope can be interpreted as an elasticity.
- What is an elasticity? A measures that relates the percentage change in  $x$  to the percentage change in  $y$ .
- For example, the slope in this example is -2.44, implying that a 1% increase in price is associated with a 2.44% *decrease* in quantity.

# OK, so when to use logs?

- Use your eyes:
  - If there are extreme values of your explanatory variable (big or small)
  - If there is a “bend” in the data
- Use your intuition:
  - When you think percentage changes matter, not changes in the level of a variable
    - Prices and quantity demanded
    - Wages/income
    - Measuring profits and productivity

# Log variables: interpretation

## 3 Types of Models:

1. Log-log:  $\log(y) = b_0 + b_1 \log(x)$

- 1% change in  $x$  is associated with a  $b_1$  percent change in  $y$
- Perhaps the most important of the three: estimates *elasticity*

2. Logs on levels:  $\log(y) = b_0 + b_1 x$

- 1 unit change in  $x$  is associated with a  $100*b_1$  percent change in  $y$

3. Levels on logs:  $y = b_0 + b_1 \log(x)$

- 1% change in  $x$  is associated with a  $b_1/100$  unit change in  $y$

# Log variables: interpretation

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# Best Practices

- Anticipate whether the association between  $y$  and  $x$  is linear.
- Plot your data.
- Don't forget about lurking variables
- Interpret the slope carefully.
- Don't compare  $r^2$  between models with different responses.

# Outline

- Modeling and Interpreting Curved Patterns
- Transformations
- Bonus Example: optimal pricing

# Bonus Example: Optimal Pricing

## Elasticity

- Elasticity: measure that relates % change in  $x$  to % change in  $y$ ; slope in a log-log regression equation.
- Can use elasticity to find the optimal price of pet food for the supermarket chain.
  - Need measure of cost (e.g. 60 cents) and price elasticity

# Logarithm Transformation

## Elasticity

$$\begin{aligned}\text{Optimal Price} &= \text{Cost} * \left( \frac{\text{Elasticity}}{\text{Elasticity} + 1} \right) \\ &= 0.60 \left( \frac{-2.44}{-2.44 + 1} \right) \\ &= \$1.017\end{aligned}$$

- Estimated profit is maximized at this price.

# Optimal Pricing

## Motivation

- How much should a convenience store charge for a half-gallon of orange juice?
- The orange juice costs the chain \$1 to stock and sell.

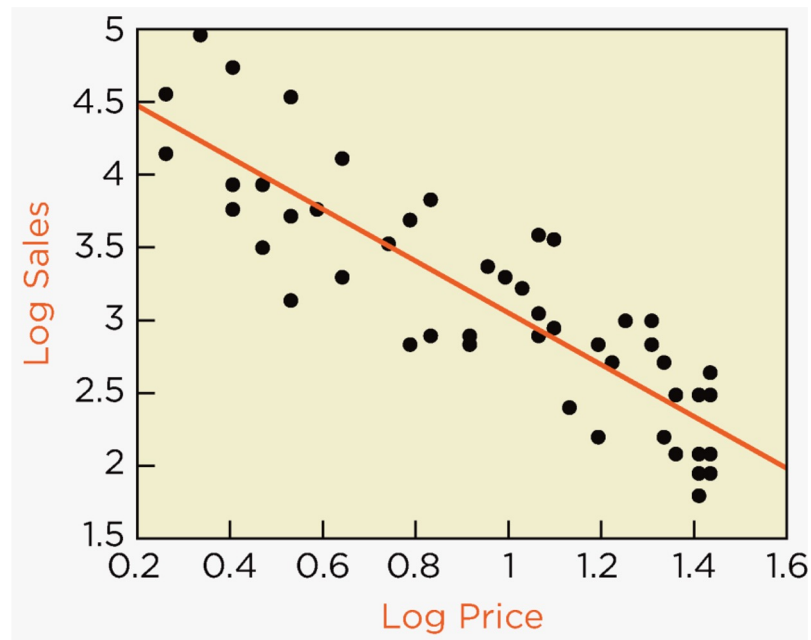
# Optimal Pricing

## Method

- In order to determine the optimal price, need to estimate elasticity from a regression of log sales on log price.
- The chain collected data on sales of orange juice from stores at 50 different locations.
- The stores sold orange juice at different prices.

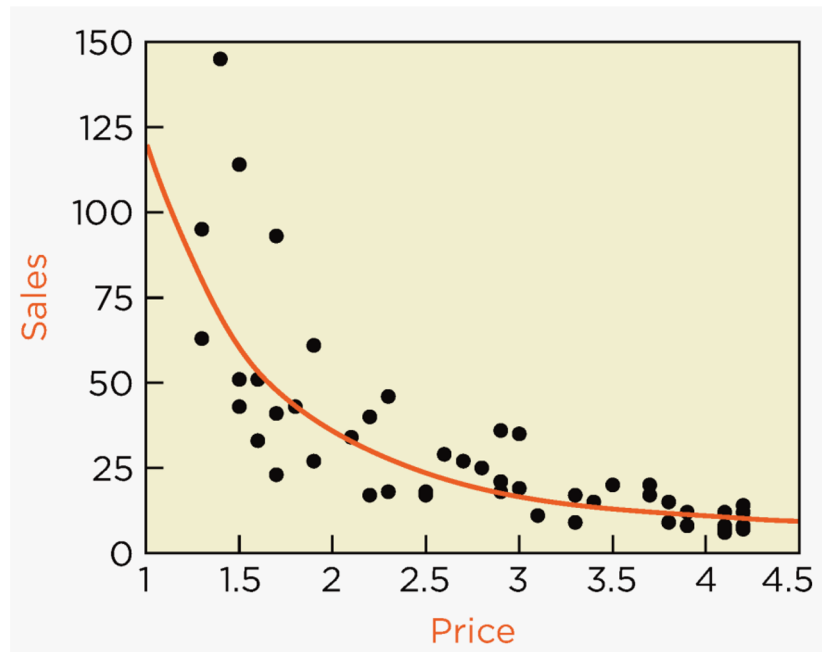
# Optimal Pricing

## Method



# Optimal Pricing

## Method



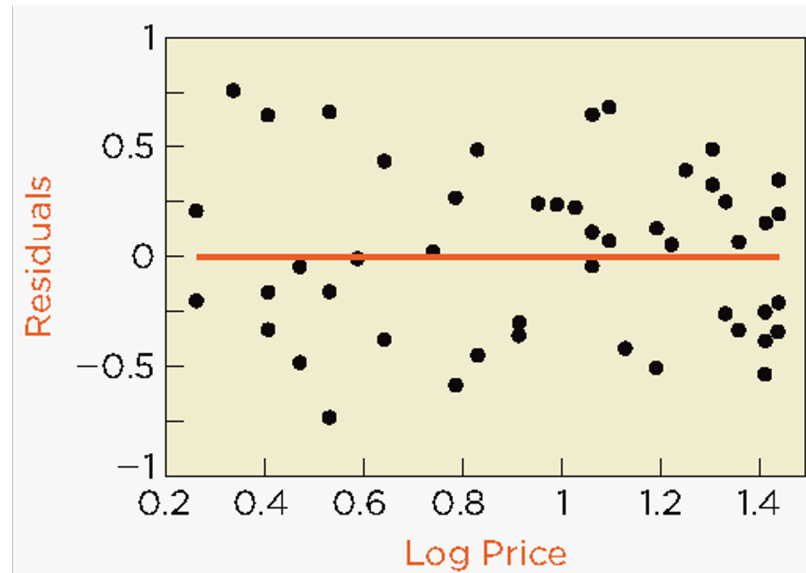
# Optimal Pricing

## Mechanics

- The least squares regression of log sales on log price is
- *Estimated log Sales* =  $4.81 - 1.75 \log \text{Price}$
- i.e. a 1% increase in price leads to a 1.75% decline in sales

# Optimal Pricing

## Mechanics – Check Conditions



- All conditions satisfied.

# Optimal Pricing

## Mechanics – Optimal Price

$$c\gamma / (1 + \gamma) = \$1(-1.75) / (1 - 1.75) = \$2.33$$

- At \$2.33, each store can expect to sell 28 cartons of orange juice for an estimated profit of \$37.24.

# Optimal Pricing

## Message

- The chain would make higher profits by decreasing the price of a half-gallon of orange juice from its current price of \$3.00 to \$2.33.